

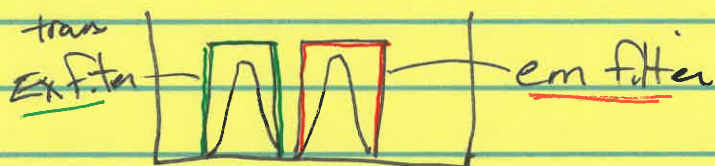
Lecture 15

2/23/26
JRA

Dichromatic Mirror

Epi Fluorescence - if not using a laser have white light source

Ex/Em profile

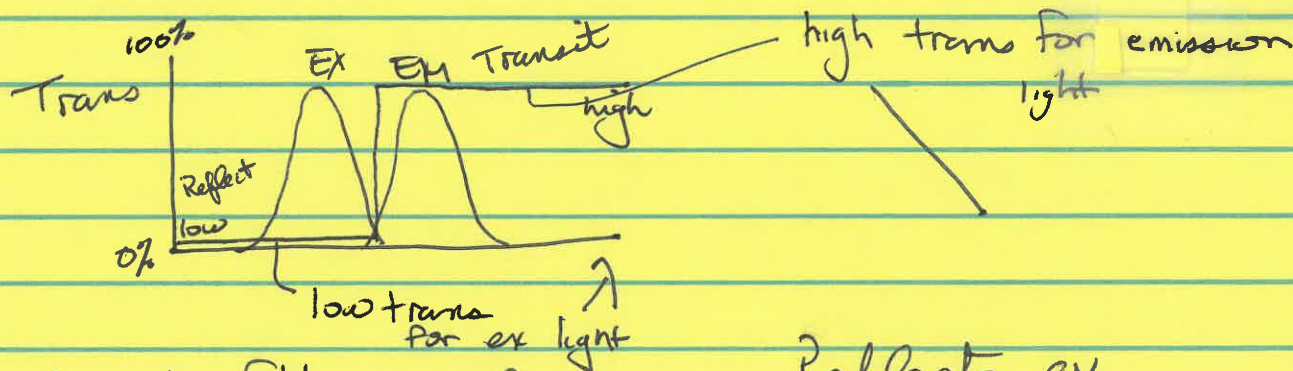


Want a filter

Ex
Em

High transmission where mol absorbs
Zero " " " does not absorb

Dichromatic mirror profile



Look at filter cube

Reflects ex & em

Reflects ex
Transmits em

HW - make this work for 2 chromophores

PP - Dichromatic mirror for three Fluorophores
Triple bandpass excitation

DAPI - reflect ex, transmit em

FITC - " " " "

Texas Red - " " " "

Always make sure low trans for ex
high " " em

Rise of (F) linked to technologies like these

Now have feel for the way (F) works

Stokes shift - use filters

Setup of epi-fluorescence microscope

Today problems & clever fixes

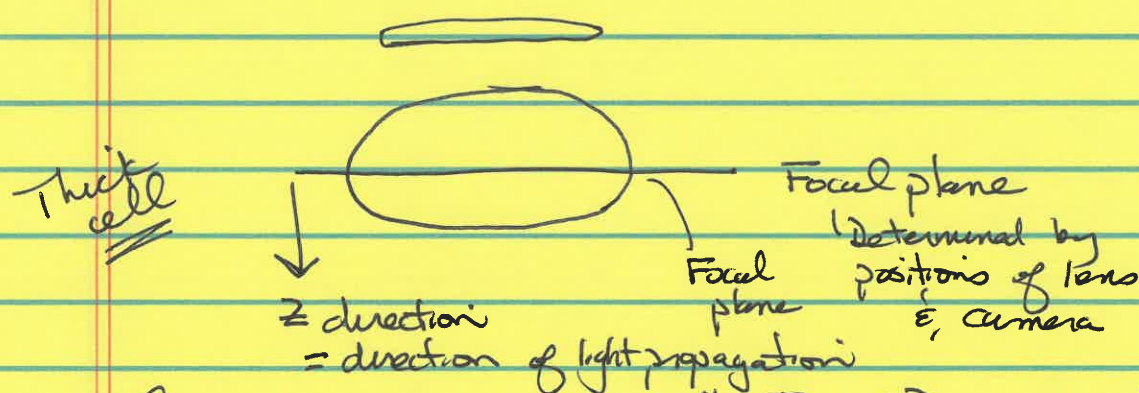
Start w/ their specimens

Multiple modes

- Single • Single color image
- Multi • Two color " & superpose mostly seq. imaging
- Time lapse

Three-dim imaging of thick samples

In many cases you want to image in 3D



Almost every cell is "thick" from viewpoint of microscopy

Camera or eye collects 2D images

Orthogonal directions are x/y

Collect z stack - move specimen, get new planes to come into focus.

Most microscopes, stage moves
Take a new picture of each plane

If you want 3D information

- 1) Take a picture of one focal plane (depth)
- 2) Move the specimen, so new plane is in focus
- 3) Take another picture
- ⋮
- n) Take pictures at all depths

Typical separation between images
 $\approx 200 \text{ nm}$ or $0.3 \mu\text{m}$

1) Take Z stack (series of 2D pictures at different depths)

4) computer will do a reconstruction

VIDEO op 5 $\frac{1}{2}$ CM Kurt Thorn

Embryonic mouse lung

Get 3D Structure

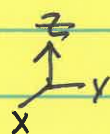
Rotate - see from different angles

Conventional

10 μm need 30 pictures 5-10 μm

Optical sectioning

move spec
 dif depth
 in focus



$\Rightarrow$ Stack of images

$\downarrow$ lower sample

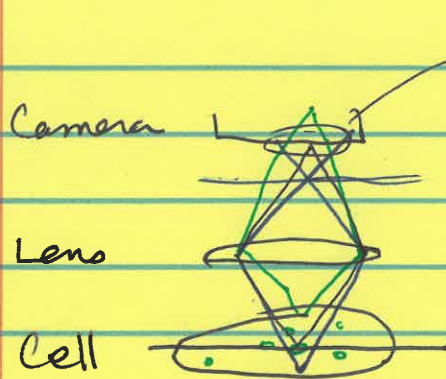
Ex C elegans GFP Ps Red

Slices 250 nm apart

Start at bottom & move up

A huge problem arises w/ standard microscopy
 "Blur problem" in 3D fluorescence microscopy

15-5



Blur spot if fluorophore too high or too low

All rays emitted by fluorophore come to focus at single point

Fluorophores throughout cell - all depths

Focal plane = plane that satisfies thin-lens eqn

Lens also picks up light from OOF fluorophores
These can focus too fast or too late
→ OOF at camera

Radiation doesn't stop

Fluorophore too high → focus too late blur

^{std} Conventional fluorescence microscopy - "wide-field"
epi-fluorescence images are very blurry
unless specimen very thin

VIDEO pt 2

Schematic of simple microscope

Ignore Tube Lens

Problem w/ conv fl microscopy - one plane in focus
 - lots of OOF light
 → haze

Simple trick to overcome - put pinhole to block light

Focal plane - goes thru

Nonfocal " - blocked, most misses pinhole (rejected)

→ Confocal because pinhole & sample simultaneously in focus

Shine laser at pt in sample, reject OOF light
 microscopy
 Laser scanning C_μ - scan over whole cell

Don't take picture all at once



→ Reconstruct

Image spot by spot & build up image

Usually use lasers to get bright enough illum

Diffraction limited spot

Change angle of illum to move across sample

Change angle light on

Looked at PP image

Do this w/ mirrors

ANIMATION